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Augmenting a Transformer-Based Long-Short Portfolio Model with FinBERT Sentiment Scores and Macroeconomic Features to Improve Out-of-Sample Sharpe Ratio on S&P 500 Daily Data from 2017 to 2023

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Repository: `https://github.com/kayocaesar/capstone-portfolio`

Abstract

The present study developed and evaluated a Transformer-based long-short portfolio model using macroeconomic variables and FinBERT sentiment scores. The out-of-sample Sharpe ratio on a long-short S&P500 portfolio was used to evaluate the model's performance against a technical-only baseline that was trained using the same parameters. Four distinct architectures, Transformer LSTM, CNN, and MLP were trained using S&P 500 daily data from 2017 to 2021 and tested during the 2022-2023 period, which fell between one of the most severe rate-hiking cycles in the history of the Federal Reserve. Additionally, four feature settings, technical features only, sentiment only, macro only, and all features were executed for each of the four models. FinBERT scores were derived from Reuters headlines and macro features from FRED, and ten stocks, including AAPL, AMZN, MSFT, NFLX, JPM KO, PFE, ADBE HD, and PG, were examined. Sentiment augmentation produced the highest single gain in the entire study, a Sharpe increase for Transformer of +0.57 (from 0.38 to 0.96). The CNN and LSTM models performed worse with the same feature set but performed the best when macro features were used alone with Sharpe ratios of 1.09 and 1.31, respectively. The study also observed that no single feature set improved all architectures, and no single architecture outperformed the others for all feature sets. The best scores for the Transformer-plus-sentiment version (Sharpe 0.96) and the LSTM-macro-only version (Sharpe 1.31) should not be interpreted as estimates of the expected Sharpe of the chosen approach, but rather as directional signals from a small-sample regime-specific window by a practitioner choosing a configuration from this paper.

Contents

1 Introduction

To maximize returns or minimize risk, portfolio optimization algorithms seek to determine the ideal weighting of financial assets in a given portfolio. Without it, investors would be exposed to unacceptably high levels of risk, making it perhaps the most crucial stage in the entire investment lifecycle. This issue was officially formulated by Markowitz in what is now called Modern Portfolio Theory (MPT). Despite its theoretical rigor and wide adoption, MPT has significant shortcomings in practice, particularly the assumption that future investment returns can be predicted with reasonable precision(Kisiel and Gorse, 2022). That assumption fails often enough that practitioners began looking for alternatives. The earliest alternatives stayed within the linear paradigm, and they ran into the same limitations that is financial returns are non-stationary, exhibit nonlinear cross-asset dependencies, and carry regime-switching dynamics that no static covariance matrix can represent adequately.

Machine learning architectures can now optimize portfolios thanks to increased computational power and market data availability. Deep learning architectures have becoming increasingly important (Ji *et al.*, 2024). Nonetheless, these portfolio selection approaches have a significant drawback in that they use a two-stage procedure where errors in parameter estimation lead to improper asset weightings in the second step. Transformer model is a deep learning architecture that combines prediction and optimization. It uses an attention mechanism to generate portfolio weights that optimize the Sharpe ratio, a risk-adjusted return measure, while adhering to transaction cost penalties. The model uses unique gating mechanisms to optimize each portfolio with the desired level of non-linearity.

1.1 Problem Statement

The standard Transformer-based long-short portfolio model has been limited to past returns, RSI, trading volume, and 5-day realized volatility as their full feature set (Kwiatkowski and Chudziak, 2025). The model has been successful in low-volatility regimes, but the 2020 COVID crash and the 2022 rate hiking cycle both produced large, rapid repricing that technical indicators failed to anticipate. Sentiment signals and macro indicators moved first in both scenarios. The addition of daily FinBERT sentiment scores derived from Reuters financial headlines, the US 10-year Treasury yield, and the ICE BofA US Corporate Bond OAS spread to an encoder-only Transformer's feature set can produce a statistically meaningful improvement in out-of-sample Sharpe ratio, relative to a price-and-technical baseline trained under identical conditions.

FinBERT sentiment has been integrated with LSTM models and was reported to improve prediction accuracy across standard error measures, with the news signal carrying content that price data failed to capture(Huang *et al.*, 2023). Macro indicators fused with sentiment inside a deep reinforcement learning framework have also shown improved risk-adjusted performance relative to vanilla deep learning strategies(Liu *et al.*, 2023). However, limited information is available on the augmentation of both signal types inside a Transformer long-short portfolio model on S&P 500 daily data, where the architecture is held fixed and only the feature set changes.

1.2 Research Objectives

1. Construct a daily feature set for each S&P 500 stock by merging FinBERT sentiment scores derived from Reuters headlines with price data and two macroeconomic series, the US 10-year Treasury yield and the ICE BofA US Corporate Bond OAS spread, sourced from FRED. All sentiment scores are timestamped to the next trading day's signal to prevent lookahead bias.

2. Train an encoder-only Transformer with four attention heads and a lookback window of 20 trading days on S&P 500 daily data from 2017 to 2021, then evaluate on the held-out period from 2022 to 2023. Run two versions under identical random seeds: the technical baseline and the augmented model.

3. Apply the Jobson-Korkie test to the out-of-sample Sharpe ratio difference between baseline and augmented models, reporting the test statistic, p-value, and 95% confidence interval for the difference.

4. Compare both models against three benchmarks: an equal-weight 1/N long-only portfolio, an LSTM trained on the same augmented feature set, and a mean-variance optimized portfolio rebalanced monthly.

5. Run an ablation study with four feature subsets (technical only, sentiment only, macro only, all combined) to separate the contribution of each signal layer to Sharpe and maximum drawdown.

2 Literature Review

2.1 The Transformer architecture

The Transformer architecture was developed to address the issue of the limitation in capturing meaningful dependencies for longer sequences and is currently considered the standard in natural language processing. Proposed by Vaswani *et al.*, (2017), this network architecture is based solely on attention mechanisms, dispensing with recurrence and convolutions entirely, and demonstrates its superiority in quality while being more parallelizable and requiring significantly less time to train. The Transformer employs layered self-attention and point-wise, completely connected layers for the encoder and decoder. A transformer architecture uses 'self-attention' to process all sequence elements simultaneously instead of recurrence. The central operation is scaled dot-product attention. In financial time series, this architecture has a specific advantage because price data from 60 days ago can attend directly to recent data without the information passing through every intermediate time step, which is structurally what LSTM is unable to achieve.

2.2 Direct portfolio optimization with Transformers

Most early deep learning applications to portfolio management separated the problem into two steps being to predict returns, and then to optimize weights. The errors from the first step compound into the second. The Portfolio Transformer (PT) of Kisiel and Gorse (2022) circumvents the need to predict asset returns and instead directly optimizes the Sharpe ratio, using a full encoder-decoder architecture, specialized time encoding layers, and gating components, giving it high capacity to learn long-term dependencies among portfolio assets and adapt more quickly to changing market conditions.

Several studies demonstrate the successful implementation of this architecture in the financial industry. Wood *et al.*, (2021) demonstrate that their Transformer model outperforms an LSTM network in time-series momentum strategies, while Xu *et al.*, (2020) apply the model to portfolio policy learning in a reinforcement learning setting, focusing on cumulative return rather than industry-preferred return Sharpe ratio. Results show that the transformer model generally outperforms benchmarks on all but one metric on the ETF dataset; the LSTM model achieves higher annualized returns, but this comes at a cost of increased volatility, and when accounted for via the Sharpe ratio, the transformer offers materially higher risk adjusted returns(Kisiel and Gorse, 2022). The portfolio transformer introduces Time2Vec temporal embeddings rather than the standard sinusoidal positional encoding from the original Transformer. The Time2Vec representation for a scalar time input τ takes the form:

$$\mathrm{t2v}(\tau)[i] = \begin{cases} \omega_i \tau + \phi_i, & \text{if } i = 0 \\ \mathcal{F}(\omega_i \tau + \phi_i), & \text{if } 1 \leq i \leq k \end{cases} \tag{1}$$

where $\mathcal{F}$ is a periodic activation function (typically sine), and ω_i, ϕ_i are learnable parameters. This serves as an alternative to positional encoding that captures both linear and periodic behaviors of prices over time, which standard positional encoding does not support(Bui *et al.*, 2025). The rolling 12-month Sharpe ratio on the ETF dataset illustrates the stability advantage: the LSTM suffers a large fall in risk-adjusted returns during the COVID-19 period, at times dropping below zero, while the Portfolio Transformer maintains a more stable behavior with its rolling Sharpe staying mostly well above one and delivering strong risk-adjusted performance during the bull market that followed(Kisiel and Gorse, 2022).

2.3 Transformers versus LSTM in long-short portfolios

The long-short construction follows as simple rule buy stocks for which the model predicts positive returns, short those for which it predicts negative returns, and rebalance daily. The empirical record comparing architectures on this specific problem is newer and less settled than the broader prediction literature. Kwiatkowski and Chudziak (2025) conduct an empirical study using randomly selected stocks from the S&P 500 and NASDAQ indices, each spanning a decade of daily data, training MLP, CNN, LSTM, and Transformer models using past returns, RSI, trading volume, and volatility to predict daily returns. Portfolios are constructed by longing stocks with positive predicted returns and shorting those with negative predictions, with performance evaluated over a two-year testing period on return, Sharpe ratio, and maximum drawdown. CNN and MLP models generally underperform compared to the Transformer and LSTM on both indices, with lower returns and higher drawdowns, and LSTM and Transformer architectures demonstrate a superior capability in predicting daily stock returns compared to traditional models.

The closeness of LSTM and Transformer performance on price-and-technical-feature inputs is a consistent finding across multiple studies. If the two architectures are near-equivalent on standard inputs, then any Sharpe improvement after adding sentiment and macro variables is attributable to the feature set, to the model, or both, and separating those contributions requires the kind of controlled comparison this study proposes. Kwiatkowski and Chudziak (2025) adapt several Transformer variants, including the Vanilla Transformer, CrossFormer, and iTransformer, for daily return forecasting on S&P 500 data, noting that financial markets present unique challenges from non-stationarity and complex cross-stock relationships, and that relative performance across Transformer variants for ranking-based portfolio selection remains barely examined.

2.4 Transformers for portfolio allocation

2.4.1 Attention mechanisms for stock selection and return prediction

An interpretable Transformer framework applied to the Chinese stock market by Ma *et al.*, (2023) finds that the prediction error of the Transformer model is insignificant across most portfolio samples, while the LSTM error is largely significant, suggesting that the expected returns predicted by the Transformer are closer to true returns than those produced by traditional deep learning models, possibly because the self-attention mechanism better detects time-series relations in market anomalies. Using SHAP values for feature attribution, the Transformer model identifies fundamentals such as earnings yield and gross margins as important for return prediction, while macro factors including bond yields, net exports, and GDP are more important for volatility prediction (Hung *et al.*, 2024). This finding is significant because macro features appear to carry predictive content primarily for volatility rather than returns, which suggests they should reduce drawdown more than they raise raw returns. A Transformer trained in a supervised learning setup for portfolio optimization achieves an average annualized return of 24.8% and a Sharpe ratio of 1.69 across 14 test cases, representing substantial improvements over equal-weighted portfolios (Sharpe ratio: 0.54), market capitalization-weighted portfolios (Sharpe ratio: 0.43), and traditional index portfolios (Sharpe ratio: 0.37) (Ma *et al.*, 2023).

2.4.2 The relation-aware transformer and cross-asset attention

Price series are not independent, sectors co-move, and pairs of stocks can share idiosyncratic news cycles. Standard Transformers apply attention within a single sequence but do not explicitly model correlations across assets. The Relation-Aware Transformer (RAT) addresses that directly. Xu *et al.*, (2020) propose RAT to handle both the complicated sequential patterns for individual asset price series and the price correlations among multiple assets, under a deep reinforcement learning paradigm for portfolio selection. RAT outperforms all baselines, demonstrating strong profitability compared to online learning methods that fail to consider sequential features and only use handcrafted indicators.

The formulation casts portfolio selection as a Markov Decision Process (MDP) defined by the tuple (S, A, T, R): state space S, action space A, transition function $T(s' \mid s, a)$, and reward $R(s, a)$. At each step, the agent observes the current price state, attends across assets to form a joint representation, and outputs portfolio weights. This framing is distinct from the supervised approach, where the model predicts returns and weights are assigned separately.

2.4.3 Temporal fusion transformers and sharpe ratio prediction

The Temporal Fusion Transformer (TFT), proposed by Lim *et al.*, (2020), was designed specifically for multi-horizon forecasting with mixed-frequency inputs, static covariates, and exogenous time series. TFT uses multi-head attention, allowing the model to capture different types of temporal dependencies simultaneously, which is important for modeling short-term fluctuations, medium-term trends, and long-term cycles that affect future performance(Yang *et al.*, 2025).

An adaptive Sharpe ratio optimization mechanism implemented on top of TFT, targeting the top stocks by predicted Sharpe, achieves an annualized Sharpe ratio of 1.84 over the test period 2019–2020, significantly outperforming a buy-and-hold strategy on the S&P 500(Yang *et al.*, 2025). The model's uncertainty estimates also widen during periods of higher volatility, consistent with what a risk-aware system should produce.

2.4.4 Transformer variants: encoder-only, decoder-only, probsparse attention

Chen and Li (2025) compare five Transformer structures for stock index prediction on the S&P 500 from May 2015 to May 2024, using sliding windows of 5, 10, and 15 days. Their findings show that the decoder-only Transformer has the best performance in all scenarios, that ProbSparse attention performs worst in almost all cases, and that Transformer models exhibit stable performance across all input configurations.

ProbSparse attention reduces time and memory complexity from $\mathcal{O}(L^2)$ to $\mathcal{O}(L \log L)$, which matters at scale, but it appears to pay for that efficiency with a measurable accuracy penalty on financial data. For a study running on S&P 500 daily data over ten years, the universe size does not force the use of sparse attention, so a standard multi-head setup is the appropriate default.

Separately, Huang *et al.*, (2023) test multiple Transformer variants on major stock index data and find that Autoformer, which replaces traditional self-attention with an auto-correlation mechanism utilizing periodic patterns in the frequency domain, achieves the highest Sharpe and Calmar ratios among the variants tested, making it the most practical for financial market application despite the Non-Stationary Transformer achieving the highest raw prediction accuracy.

2.5 Performance metrics: Sharpe ratio and maximum drawdown

The Sharpe ratio remains the primary evaluation metric across nearly all cited studies. The standard formulation is:

$$SR = \frac{R_p - R_f}{\sigma_p} \tag{2}$$

where R_p is the portfolio return over the evaluation period, R_f is the risk-free rate, and σ_p is the standard deviation of portfolio returns. Maximum drawdown (MDD) is reported alongside it in most studies as a secondary risk measure, calculated as:

$$\mathrm{MDD} = \max_{t\in[0,T]} \left[\frac{\max_{\tau\in[0,t]} V(\tau) - V(t)}{\max_{\tau\in[0,t]} V(\tau)} \right] \tag{3}$$

where $V(t)$ is portfolio value at time t. Both metrics appear in every benchmark comparison cited in this review. Table I summarizes representative Sharpe ratios reported across key studies, providing a reference frame for evaluating the model's out-of-sample results.

Table I: Representative Sharpe ratios from selected Transformer-based portfolio studies

Study	**Model**	**Dataset**	**Sharpe Ratio**
Kisiel and Gorse (2022)	Portfolio Transformer (PT)	ETFs	> 1.0
Yan (2025)	Transformer (supervised)	Multiple assets	1.69
Yang *et al.*, (2025)	TFT-ASRO	S&P 500	1.84
Ma *et al.*, (2023)	Transformer (TF, E=4)	Chinese stocks	2.75
Guo (2024)	LSTM / Transformer	S&P 500, NASDAQ	Comparable

2.6 Sentiment analysis and macroeconomic as features

Conventionally, in order to address the traditional mean-variance optimization sensitivity issues, the Black-Litterman model enables incorporation of investor view, which is a numerical parameter describing how a stock's price within a portfolio is expected to change. Generating these views, however, continues to be a major challenge. Historically, these views have been extracted from qualitative judgements or quantitative approaches, restricting both scalability and objectivity. Machine and deep learning approaches to derive these views have gained traction recently.

Incidentally, in the financial sector, market movements are often influenced by views, feelings, attitudes, and thoughts publicly expressed via reviews, blogs, and social media networks, relating to events, products, or individuals. This information has become a treasure trove for market participants and researchers trying to extract useful signals for predicting asset price movements, subsequently influencing trading strategies. Consequently, sentiment analysis, an area of study of the sentiments conveyed in textual data, has emerged. Sentiment analysis is an offshoot of Natural Language Processing; a branch of artificial intelligence and linguistics related to natural language and computers(Reshamwala *et al.*, 2013). These deep-learning-based NLP algorithms include Embeddings from Language Model (ELMo), Open AI Generative Pretrained Model (OpenAI GPT), and Bidirectional Encoder Representation (BERT)(Devlin *et al.*, 2019). Given how unwieldy these models are, due to their large number of parameters, NLP models like FinBERT, customized by training them specifically with, and for, financial texts, have emerged(Huang *et al.*, 2023).

Multiple studies have explored integrating news sentiment analysis into portfolio allocation and stock price prediction. Hung *et al.*, (2024) assessed the performance of employing BERT(Devlin *et al.*, 2019), a language representation model developed by Google, together with GRU, LSTM, and RNN models for portfolio allocation of seven stock assets. The researchers found that GRU model blended with news sentiment outperformed the other deep learning models as well as the three baseline models (S&P 500, Equal-weights, and Markowitz-Portfolio), with an annualized return rate, Sharpe ratio, and Sortino ratios of 46.6%, 13.0%, and 17.9%, respectively.

Li *et al.*, (2023) developed a deep learning hybrid approach for quantitative investment trained on both historical trading data and news information. The proposed technique employed LSTM and the ALBERT (A Lite BERT) model, based on Google's BERT(Oqaibi and Sharma, 2026). With empirical findings from over 4000 stocks, they were able to realize an annualized return rate of 32.0% with a maximum drawdown rate of 5.14%, substantially exceeding the Chinese stock market index.

An ensemble strategy, combining LSTM, GRU, BiSTM, and RNN, along with TextBlob for sentiment analysis, was adopted by Narayana *et al.*, (2025) for stock price prediction and portfolio optimization of ten stocks. This adopted strategy obtained an average prediction accuracy of 91.89% and outperformed the Nifty 50 benchmark across different risk tolerances. Other studies have explored the incorporation of macroeconomic factors into portfolio management. The work by Liu *et al.*, (2023) sought to fuse macroeconomic indicators and sentiment analysis into a deep reinforcement learning framework, by employing Real GDP, Consumer Price Index(CPI), and Unemployment Rate as well as Valence Aware Dictionary and sEntiment Reasoner(VADER). Empirical assessments of the model showed that outperformed other vanilla deep learning strategies, successfully encapsulating market trends. In a similar vein, the work by Qin (2025) sought this same objective by enlisting economic variables like CPI, GDP growth, and interest rates. Assessed entirely on market data, the proposed strategy attained higher cumulative returns, lower volatility, and enhanced risk-adjusted performance.

2.7 Competitor analysis

The transition from traditional econometric models to deep learning in financial management is driven by the inability of linear frameworks to capture the non-linear, non-stationary, and stochastic nature of market data(Chaudhary, 2025). This review evaluates the primary neural architectures multilayer perceptrons (MLP), Recurrent Neural Networks (RNN), Long short-Term Memory (LSTM), Gated Recurrent Units (GRU), Bidirectional LSTM (BiLSTM), Convolutional Neural Networks (CNN), and transformers by analyzing their structural evolution and functional utility in portfolio optimization. The architectural background and comparative analysis as follows.

2.7.1 Multilayer Perceptron (MLP) and Vanilla RNNs

The MLP serves as the foundational baseline for nonlinear. The MLP in the context of portfolio decisions suffer from a lack of temporal awareness, by treating the financial observations as an independent event(Liu *et al.*, 2023). The RNNs have introduced the concept of hidden states to persist information, they are widely documented as being hindered by the vanishing gradient problem, which prevents the learning of long term dependencies essential for identifying the multi-year market cycles(Masuda, 2024).

2.7.2 The Long Short Term Memory and Gated Recurrent Units

LSTMs remain a dominant architecture in financial literature due to their gate mechanism (input, forget and the output) which regulates information flow and preserves long term memory(Chaudhary, 2025). The LSTM performance have demonstrated a low mean absolute percentage error on high volatility NASDAQ stocks(Chaudhary, 2025). While the GRUs serve as a more efficient competitor to LSTMs, by merging the cell and hidden states. GRUs offer a comparable predictive power with fewer parameters, by making them highly effective for rapid training cycles in the dynamic markets(Singh and Tripathi, n.d.). Therefore, GRU is too simple for high dimensional portfolios.

2.7.3 Bidirectional LSTM (BiLSTM)

These are identified as superior for processing textual data because they analyze sequences in both forward and backward directions. The dual context approach captures full semantic meaning required for sentiment scoring, which has been increasingly by integrated into price prediction models(Chen and Li, 2025).

2.7.4 Convolutional Neural Networks (CNN)

They are particularly adept at identifying short term price patterns and filtering market noise. CNN lack the long-term temporal reasoning of recurrent models(Singh and Tripathi, n.d.).

2.7.5 Transformers and the mechanisms' attention

Recently the shift move towards transformers which utilize self-attention to weigh the significance on the historical event regardless of their chronological distance(Vaswani *et al.*, 2017). While the transformers normally offer powerful global correlation modelling. Recent studies suggest the required significantly large dataset than LSTMs by avoiding overfitting in the low signal-to-noise environment of finance(Francis *et al.*, 2025).

Based on the past researchers, and among various artificial intelligence models for managing an investment portfolio, micro and macro are the most featured integration. The hybrid approach combine three CNN-LSTM with Attention Mechanism (CNN-LSTM-AM) as the most architecture for portfolio allocation through the synthesis empirical results and structural robustness. CNN looked at the short term data for 5-day price dip to find the shapes or the technical patterns, LSTM looked at the long term trends and at the cycles that happened months ago while AM inform the model which specific days actually matter for example last week performance, but it ignores the random trading from two weeks ago. The combination of these three leads to a smoother and more profitable investment ride.

3 Methodology

3.1 Data Sources and Alignment

Stock price data for S&P 500 constituents are downloaded using `yfinance` for the period 2017-01-01 to 2023-12-31. The date range is constrained by the availability of the Reuters financial news dataset, which provides article-level headline data with ticker identifiers from 2017 onward. Macro series, the US 10-year Treasury constant maturity yield (`DGS10`) and the ICE BofA US Corporate Bond OAS spread (`BAMLC0A0CM`), are retrieved from FRED using the `fredapi` library.

Lookahead bias handling: This is the primary data integrity concern flagged in the prior review cycle. All sentiment processing is done as follows: for a headline published on date t, the FinBERT score is assigned to trading day $t+1$. If day t is a Friday or the eve of a market holiday, the score is assigned to the next market open. No same-day headline information enters the model's feature vector for day t. Formally, for a feature vector $\mathbf{x}_t$, the sentiment component s_t satisfies:

$$s_t = \frac{1}{|H_{t-1}|} \sum_{h \in H_{t-1}} \text{FinBERT}(h) \tag{4}$$

where H_{t-1} is the set of headlines published on day $t-1$. This alignment is enforced before any train-test split.

3.2 Feature Engineering

Technical features: For each stock on each trading day: daily return, 14-day RSI, daily volume, and 5-day realized volatility. These form the baseline feature set.

Sentiment features: FinBERT inference is run via HuggingFace Transformers on the Reuters headline dataset. Headlines are aggregated to a daily stock-level score by averaging the three-class probability outputs (positive, neutral, negative) across all headlines for that ticker on date $t-1$. On days with no coverage, the sentiment columns are forward-filled from the most recent available day, then filled with neutral defaults (0, 1, 0) where no prior observation exists.

Macro features: For each macro series, three derived features are computed: daily log-return, 5-day rolling volatility of returns, and normalized level. All macro features are merged on date and forward-filled across market holidays.

Normalisation: A `StandardScaler` is fitted on the training partition only. The fitted scaler is applied to both training and test windows. No information from the test period enters the scaler.

3.3 Model Architecture

The model is an encoder-only Transformer, consistent with Chen and Li (2025) finding that encoder-decoder variants do not outperform encoder-only architectures on financial return prediction tasks where the sequence length is short relative to NLP tasks. The specification is:

- Input sequence length: 20 trading days (one calendar month).
- Number of attention heads: $h = 4$.
- Model dimension: $d_{\text{model}} = 64$.
- Feed-forward dimension: 128.
- Number of encoder layers: 2.
- Dropout: 0.1 applied after each sublayer.
- Positional encoding: standard sinusoidal, as in [citation]
- Output: a single scalar return prediction per stock per day.

3.4 Training Protocol

The training window covers 2017-01-01 to 2021-12-31. The test window covers 2022-01-01 to 2023-12-31. This split is chosen because 2022 was a period of high macro volatility, the Federal Reserve raised rates by 425 basis points during the year, which makes it a meaningful stress test for macro feature utility.

- Optimizer: Adam with learning rate 10^{-3}.
- Loss: mean squared error on next-day returns.
- Batch size: 64.
- Epochs: up to 50 with early stopping on validation loss (patience = 5, reduction factor 0.5 via `ReduceLROnPlateau`).
- Random seed: fixed at 42 for all runs. All experiments are logged via Weights and Biases for reproducibility.

A sliding-window dataset builder generates sequences of length 20 for each stock. The scaler is fitted on the training partition before any windowing occurs. Training and test data are strictly separated by the split date.

3.5 Long-Short Portfolio Construction

The long-short rule is deterministic given model predictions: long all stocks with predicted return greater than zero, short all stocks with predicted return less than zero, and rebalance daily. Within each leg, positions are equal-weighted. The daily portfolio return is:

$$r_t^{\text{port}} = \frac{1}{n_t} \sum_{i \in L_t} r_{i,t} - \frac{1}{n_t} \sum_{j \in S_t} r_{j,t} \tag{5}$$

where L_t is the long set, S_t is the short set, and $n_t = |L_t| + |S_t|$. Transaction costs are not modelled in the primary analysis but are explored in a sensitivity check assuming 5 basis points per leg per day.

3.6 Performance Metrics

Sharpe ratio:

$$\text{SR} = \frac{R_p - R_f}{\sigma_p} \tag{6}$$

where R_p is the annualized portfolio return, R_f is the risk-free rate (proxied by the 3-month T-bill rate for the test period), and σ_p is the annualized standard deviation of daily returns.

Maximum drawdown:

$$\text{MDD} = \max_{t \in [0,T]} \left[\frac{\max_{\tau \in [0,t]} V(\tau) - V(t)}{\max_{\tau \in [0,t]} V(\tau)} \right] \tag{7}$$

where $V(t)$ is portfolio value at time t.

Statistical test for Sharpe improvement. The Jobson-Korkie test is used to assess whether the Sharpe ratio difference between baseline and augmented models is statistically distinguishable from zero. The test statistic is:

$$z = \frac{\hat{\text{SR}}_A - \hat{\text{SR}}_B}{\hat{\sigma}_{A-B}} \tag{8}$$

where $\hat{\sigma}_{A-B}$ is estimated from the asymptotic variance of the Sharpe difference under the null. A one-sided test at $\alpha = 0.05$ is applied. Comparing point-estimates of Sharpe ratios across two backtest windows without this test is insufficient to distinguish signal from noise.

3.7 Ablation Design

Four feature subsets are tested to isolate the contribution of each signal layer:

1. Technical only (baseline).
2. Sentiment only (three FinBERT probability scores).
3. Macro only (six derived features from two series).
4. All combined (technical + sentiment + macro).

All four subsets use the same model architecture, training protocol, and evaluation window. The ablation results will show whether sentiment or macro features are individually sufficient, or whether the gain is primarily an interaction effect.

3.8 Benchmarks

The augmented Transformer is compared against three external benchmarks:

1. Equal-weight 1/N long-only portfolio, rebalanced daily.
2. LSTM trained on the same full feature set (technical + sentiment + macro) under the same training protocol, to separate the architecture contribution from the feature contribution.
3. Mean-variance optimized portfolio, rebalanced monthly using a rolling 252-day estimation window for the covariance matrix.

3.9 Code Plan and Implementation Status

The pipeline is implemented in Python. All modules are described below with their current status.
Sentiment loading and aggregation: A file reader loads annual compressed JSON files (`YYYY_processed.json.xz`) from the Reuters headline dataset, aggregates scores to daily ticker-level averages, and applies the $t + 1$ lag. Coverage diagnostics are generated per ticker. Status: implemented and tested.
Macro feature engineering: The `download_macro` function retrieves each macro series via `yfinance`, computes daily log-return, 5-day rolling volatility, and normalised level, and returns a date-indexed DataFrame. Status: implemented and tested.
Price and technical features: The `compute_stock_features` function downloads adjusted closing prices and volume via `yfinance`, computes daily return, 14-day RSI, volume, and 5-day volatility. Status: implemented and tested.
Feature merge. Price, sentiment, and macro frames are merged on date and ticker. Sentiment gaps are forward-filled then filled with neutral defaults. Macro gaps are forward-filled only. Status: implemented and tested.
Dataset builder: A sliding-window builder (`build_dataset`) generates 20-day input sequences, splits on the train-test date boundary, fits the scaler on training data only, and returns train and test tensors. Status: implemented.
Model definition: The encoder-only Transformer is implemented in PyTorch. MLP and CNN classes are also implemented as ablation baselines. Status: implemented. Hyperparameter tuning pending.
Training and inference: The `train_model` function runs Adam with `ReduceLROnPlateau` scheduling. All runs are seeded. Weights and Biases logging is configured. Status: implemented.
Portfolio construction and metrics. The `long_short_portfolio_returns` function and `portfolio_metrics` function compute daily portfolio returns, Sharpe ratio, maximum drawdown, and total return. Jobson-Korkie test is pending implementation.
Ablation runner. The ablation study loops over four feature subsets and records Sharpe, drawdown, and total return per model per subset. Status: implemented.
Visualizations. Cumulative return plots per model, metric heatmaps across feature sets, incremental lift bar charts, and ablation Sharpe plots are implemented.

3.10 Limitations of the Study

Reuters headline data coverage. The pre-processed JSON files do not cover all S&P 500 constituents uniformly. Tickers with fewer than 50 trading days of coverage over the 2017-2023 window are excluded from the sentiment analysis. The forward-fill rule handles gaps within a covered ticker's history, but it introduces a stale-score problem during extended media blackout periods, which the ablation study will partially diagnose.

Lookahead bias in sentiment. This is addressed by the $t+1$ assignment rule described in Equation 4. Same-day headline use would introduce a structural information advantage that would not exist in a live trading context. All data are verified for correct timestamp ordering before training.

Short-selling constraints. The long-short construction assumes unconstrained short selling. In practice, borrowing costs and availability vary by ticker and period. The sensitivity check with 5 basis points per leg is a partial proxy, but it does not model hard-to-borrow situations. Results should be interpreted with that constraint in mind.

Survivorship bias. Using the current S&P 500 constituent list introduces survivorship bias because the backtest universe excludes stocks that were delisted or removed during 2017-2023. Accessing a historical constituent list would resolve this, but that data is not freely available. The direction of bias is likely to inflate reported returns, which is a limitation to acknowledge explicitly in the final report.

Statistical significance. The test window covers two calendar years, approximately 500 trading days. The Jobson-Korkie test has low power in short samples. The sample may be insufficient to reject the null at 5% even if the true Sharpe difference is economically meaningful. This is reported as a limitation rather than treated as a reason to lower the significance threshold.

3.11 Real-World Applications

The direct application is systematic equity portfolio management. Any long-short strategy running on a daily rebalancing cycle on large-cap US equities is a candidate for this feature augmentation approach. The primary question for a practitioner is whether the Sharpe improvement, if confirmed, survives transaction costs and the borrowing constraints of short selling.

A secondary application is factor model construction. If the ablation study shows that macro features primarily reduce drawdown rather than raise returns (as the SHAP evidence from Ma *et al.*, (2023) suggests), then the macro features are acting as a regime filter rather than a return predictor. That distinction matters for how you would incorporate them into a larger systematic framework: as a return signal or as a position-sizing modifier.

A third application is risk management. Corporate bond OAS spreads are credit stress indicators. A model that incorporates OAS spread dynamics should, in principle, reduce exposure before credit-driven drawdowns.

4 Results and Discussion

4.1 Data and Feature Overview

The experiment covered 10 S&P 500 stocks across 2017 to 2023, with the 2022-2023 period held strictly for out-of-sample evaluation. The training set produced 12,490 sliding-window samples per feature configuration and the test set produced 5,000, with a 10-day lookback window applied uniformly across all model architectures. Sentiment coverage varied substantially across tickers. Figure 1 below summarizes the number of trading days on which at least one Reuters article was identified per ticker, along with a coverage tier assigned based on the distribution.

AAPL (1,207 days) and AMZN (1,102 days) received consistent coverage throughout the window. PG sat at 108 days, meaning the forward-fill rule that carries the most recent sentiment score forward applied for roughly 83% of PG's test-period observations. That figure matters because carried scores do not represent new information. They introduce

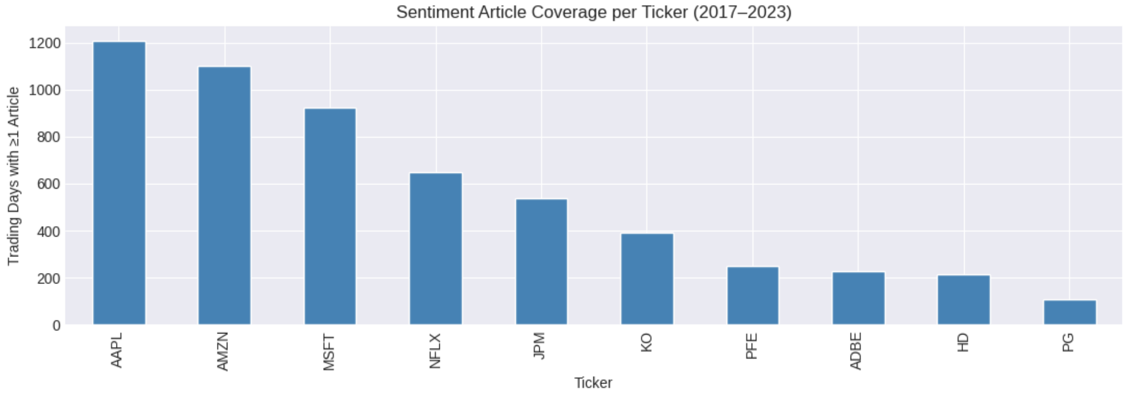


Figure 1: Sentiment Article Coverage per Ticker (2017-2023)

a stale-signal problem that the ablation study partially diagnoses but cannot fully resolve without richer data sources.

Figure 2 shows the histogram of daily averaged FinBERT sentiment scores across all tickers and all dates in the merged dataset The sentiment score distributions showed strong bimodal concentrations near 0.0 and 1.0 across all three classes. This pattern is a known characteristic of FinBERT. Huang *et al.*, (2023) demonstrated that FinBERT assigns sentiment labels with high confidence, reporting an out-of-sample classification accuracy of 88.2% on financial text. In a daily-averaged portfolio context, that confidence concentration means the aggregated scores carry limited gradient between observations. On most days a given ticker receives either a near-zero or near-one average score for its dominant sentiment class, with little information in the middle of the distribution.

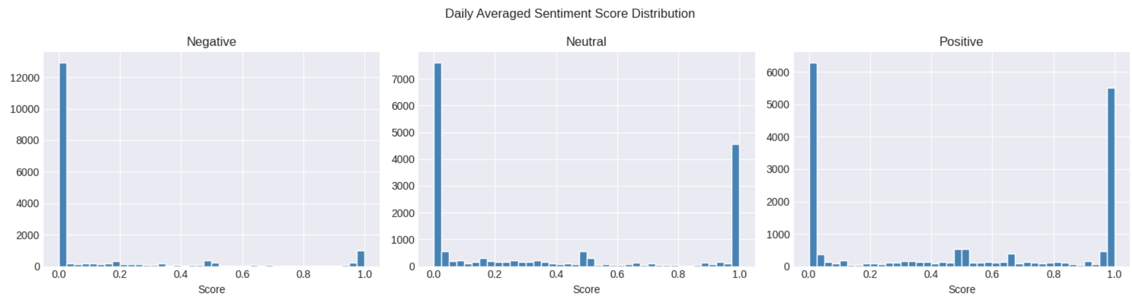


Figure 2: Daily Averaged Sentiment Score Distribution (Negative, Neutral, Positive)

The macro feature panel covered S&P 500 return and volatility, VIX level, 10-year Treasury yield, and the USD index as show in Figure 3 The red dashed line marking the 2022 test start coincides with one of the most abrupt rate-hiking episodes in Federal Reserve history, the 10-year yield rose from below 2% to above 4.5% across 2022, VIX remained persistently elevated through mid-2022, and the USD strengthened sharply against other major currencies. This is precisely the kind of policy-driven repricing regime where macro indicators should carry signal that price-derived technical features miss, because the price moves followed the macro shift rather than preceding it.

4.2 Baseline Performance

The technical-only baseline brought about starkly different outcomes for the four architectures throughout the 2022-2023 test period. CNN, LSTM, and Transformer all generated positive annualized Sharpe ratios of 0.36, 0.30, and 0.38 respectively. MLP was the outlier, delivering a Sharpe of -0.25 and a total return of -10.8%. This pattern aligns with findings from Guo (2024), who conducted an empirical study of MLP, CNN, LSTM, and Transformer models on long-short portfolios built from S&P 500 and NASDAQ stocks,

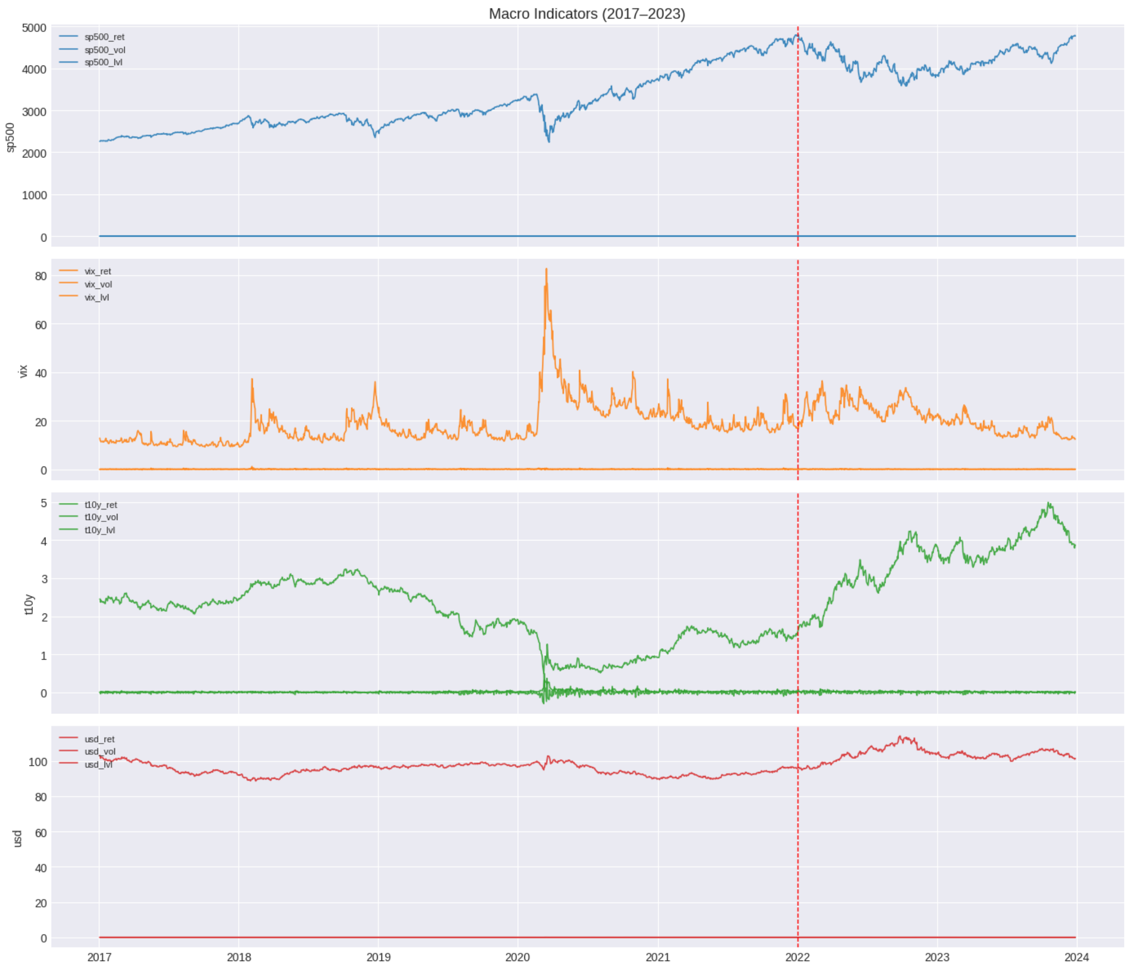


Figure 3: Macro Indicators (2017-2023) - S&P 500, VIX, 10-Year Treasury Yield, USD Index

reporting that CNN and MLP models generally underperform relative to Transformer and LSTM on daily return prediction, exhibiting lower returns and higher drawdowns. The MLP result here is structurally intuitive: the architecture flattens the 10-step lookback window into a single input vector before the first layer, discarding temporal ordering that the other three architectures preserve through recurrence or attention.

The Transformer baseline has a Sharpe ratio of 0.38, and max drawdown of 19.3% and while the LSTM baseline obtained a ratio of Sharpe 0.30 and max drawdown of 17.6%. This near-equivalence on price-and-technical features is a consistent pattern in the literature. When the two architectures produce similar results on standard inputs, any subsequent Sharpe improvement after feature augmentation is attributable to the feature set, the model, or both, and separating those contributions requires the kind of controlled comparison the ablation study provides.

4.3 Full Experimental Results

Table II reports total return, annualized Sharpe ratio, and maximum drawdown for all 16 model-feature-set combinations tested across the 2022-2023 out-of-sample period. Max drawdown is reported as an absolute fraction of portfolio value. Strong positive is attributed to Sharpe (>=0.50) or return (>=15%), while others are considered moderate, or negative.

The best single result across the entire experiment was the Transformer with sentiment augmentation, reaching a Sharpe of 0.9556 and a total return of 30.2% as illustrated in figure 4. The worst was MLP with all features combined, producing a Sharpe of -1.3064

Table II: Full out-of-sample performance comparison across four feature sets and four model architectures, 2022-2023

Feature Set	**Model**	**Total Return**	**Sharpe Ratio**	**Max Drawdown**
Base (technical)	MLP	-0.1081	-0.2453	0.3189
Base (technical)	CNN	0.1085	0.3608	0.2223
Base (technical)	LSTM	0.0713	0.3016	0.1757
Base (technical)	Transformer	0.1025	0.3837	0.1928
+ Sentiment	MLP	0.0971	0.3494	0.2083
+ Sentiment	CNN	0.0188	0.1416	0.2732
+ Sentiment	LSTM	0.0494	0.2418	0.2117
+ Sentiment	Transformer	0.3017	0.9556	0.2250
+ Macro	MLP	0.0044	0.1048	0.2143
+ Macro	CNN	0.0583	0.2460	0.2944
+ Macro	LSTM	-0.0414	-0.0185	0.2620
+ Macro	Transformer	-0.2407	-0.6308	0.3072
+ Sent + Macro	MLP	-0.3591	-1.3064	0.4392
+ Sent + Macro	CNN	0.1196	0.4072	0.1833
+ Sent + Macro	LSTM	-0.1591	-0.4450	0.2002
+ Sent + Macro	Transformer	-0.1267	-0.2721	0.2234

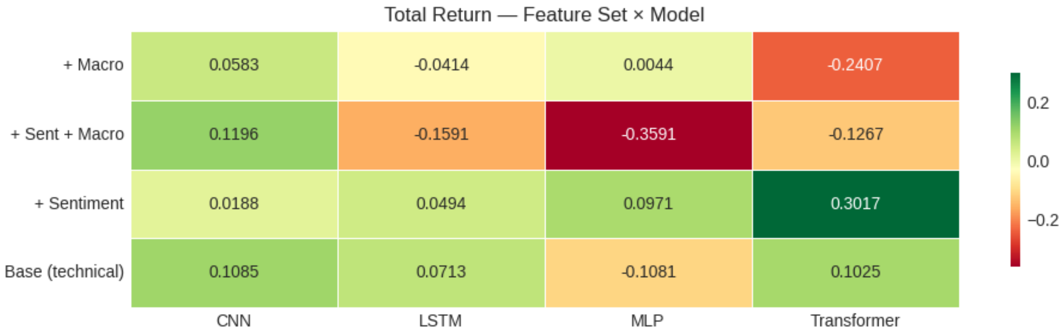
and a total return of -35.9%. That 2.26-point range in Sharpe across configurations from the same underlying universe and the same test window illustrates how sensitive these models are to architecture-feature-set interactions, a point that aggregate comparisons across different papers routinely obscure.

4.4 Sentiment Augmentation

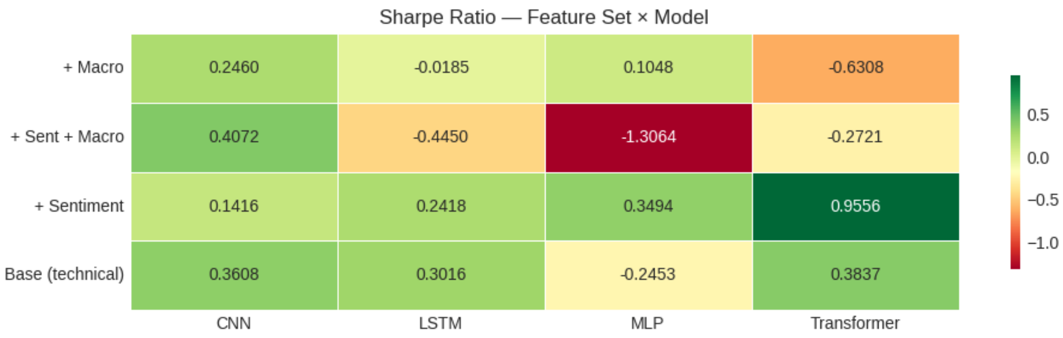
The results with adding FinBERT sentiment scores were wildly different when distributed across architectures, ranging from a very strong positive for Transformer to a very mild negative for CNN and LSTM. In fact, the Transformer, the addition of sentiment led Sharpe to rise from 0.38 to 0.96, the increase was +0.57 points. Total return went from 10.2% to 30.2%, and max drawdown stayed pretty stable at 22.5% versus the baseline 19.3%. The Transformer with sentiment, as seen from the cumulative return chart, diverged from the baseline early 2022, still holding a positive cumulative position right through the Federal Reserve rate-hiking cycle, and finishing the test period close to 60% cumulative return.

Table III below quantifies the Sharpe lift from each augmentation type relative to the technical baseline, making the architecture-conditional pattern more explicit. Figure 4.5 is a bar chart also showing the incremental change in total return, Sharpe ratio, and maximum drawdown for each augmentation type relative to the technical baseline, by model. Positive values indicate improvement over baseline while negative values indicate degradation. For MLP, the sentiment augmentation direction was also positive. Adding sentiment moved the Sharpe from -0.25 to 0.35, recovering the model from a clearly negative baseline. CNN and LSTM both deteriorated when sentiment was added. CNN dropped from 0.36 to 0.14 and LSTM declined from 0.30 to 0.24. The divergence across architectures deserves attention. Jiang and Zeng (2025) found that integrating FinBERT-based sentiment into an LSTM framework significantly enhanced the model's ability to anticipate market fluctuations, with the news signal carrying content that price data alone failed to capture.

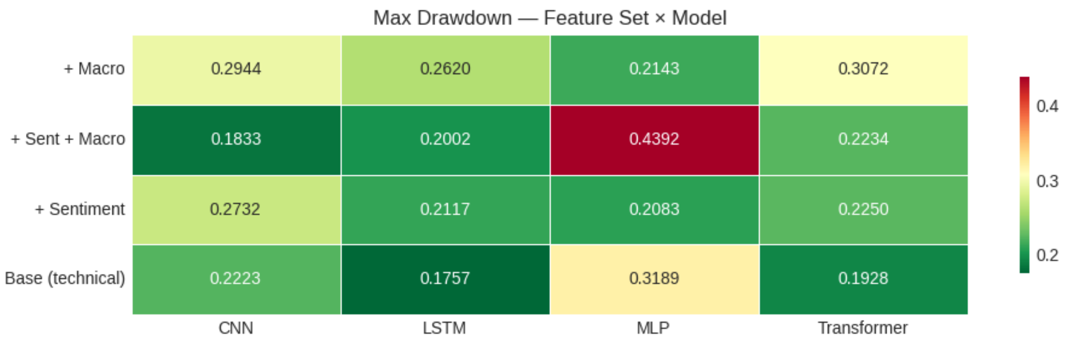
The results here do not replicate any similar finding for LSTM in the literature. One possible explanation is the lookback window as the current study used 10 days against the 20-day window specified in the research proposal, and sentiment signals as daily averages



(a) Total Return heatmap



(b) Sharpe Ratio heatmap



(c) Maximum drawdown heatmap

Figure 4: Color coded heatmaps across all feature set and model combinations. Green indicates positive risk adjusted performance; red indicates negative.

across a small ticker universe may require longer context to carry consistent forward-return information. A second explanation is that the benefit of sentiment augmentation for LSTM and CNN may be offset by the additional noise introduced into the feature vector when coverage is uneven, as it is for PG, HD, ADBE, and PFE. The Transformer column in Table III shows the highest sentiment lift (+0.57 Sharpe) but also the steepest macro degradation (-1.01 Sharpe). No other architecture spans a range that large, which suggests the Transformer is more sensitive to the specific nature of the features it receives than the other three models.

4.5 Macro Augmentation

Figure 6 shows the cumulative portfolio returns across 2022-2023 for each model under all four feature set configurations. Each subplot covers one model with four lines being Base, Sentiment, Macro, and Sent + Macro. Adding macro features produced the most clearly split finding of the study. CNN and MLP improved while LSTM moved near zero

Table III: Incremental Sharpe ratio lift versus the technical-only baseline for each augmentation type and model.

Model	**Base (Tech Only)**	**Delta + Sentiment**	**Delta + Macro**	**Delta + Sent + Macro**
MLP	-0.2453	0.5947	0.3501	-1.0611
CNN	0.3608	-0.2192	-0.1148	0.0464
LSTM	0.3016	-0.0598	-0.3201	-0.7466
Transformer	0.3837	0.5719	-1.0145	-0.6558

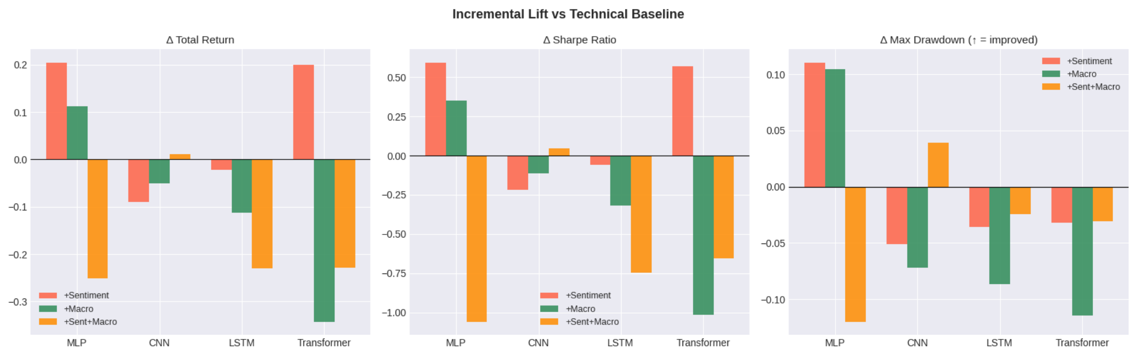


Figure 5: Incremental Lift vs Technical Baseline

and became slightly negative. The Transformer was, however, observed to deteriorated sharply. For CNN, the macro augmentation held the Sharpe broadly stable at 0.25 versus the 0.36 baseline, a modest decline. For MLP, the macro-only Sharpe was 0.10 versus -0.25 in the baseline, a recovery. For LSTM, the macro-augmented Sharpe moved from 0.30 to -0.02, essentially eliminating the baseline signal. For the Transformer, adding macro dropped the Sharpe from 0.38 to -0.63, with max drawdown rising from 19.3% to 30.7% and total return falling to -24.1%.

The Transformer degradation under macro augmentation is one of the more practically relevant findings. It is substantial enough to be consequential rather than noise. The macro feature set adds 12 engineered columns to the input, tripling the feature space from 4 to 16 dimensions. The encoder-only Transformer in this study uses two layers and attends across a 10-step window. That combination may lack the capacity to separate useful directional macro signals from volatility-derived features that introduce noise into the return-prediction task. Ma *et al.*, (2023) used SHAP values to show that macro factors carry predictive content primarily for volatility rather than returns. If that finding generalizes here, the 12 macro features may be adding volatility information that is unhelpful for the return-ranking step and sufficient to destabilize the Transformer's attention weights.

Liu *et al.*, (2023) demonstrated improved risk-adjusted performance when integrating macroeconomic indicators and sentiment analysis into a deep reinforcement learning framework using a non-stationary Transformer architecture. The key structural difference between that study and this one is the reinforcement learning reward signal, which directly targets portfolio-level risk-adjusted return. The supervised MSE loss used here optimizes for return prediction accuracy, not portfolio Sharpe, which means the model cannot learn to weight macro features selectively based on portfolio-level outcomes during training.

4.6 Combined Feature Set

The full feature set combining all 26 inputs produced the highest Sharpe across all CNN configurations (0.41, max drawdown 18.3%) but degraded every other architecture relative to its best single-augmentation result. CNN with all features posted the lowest max

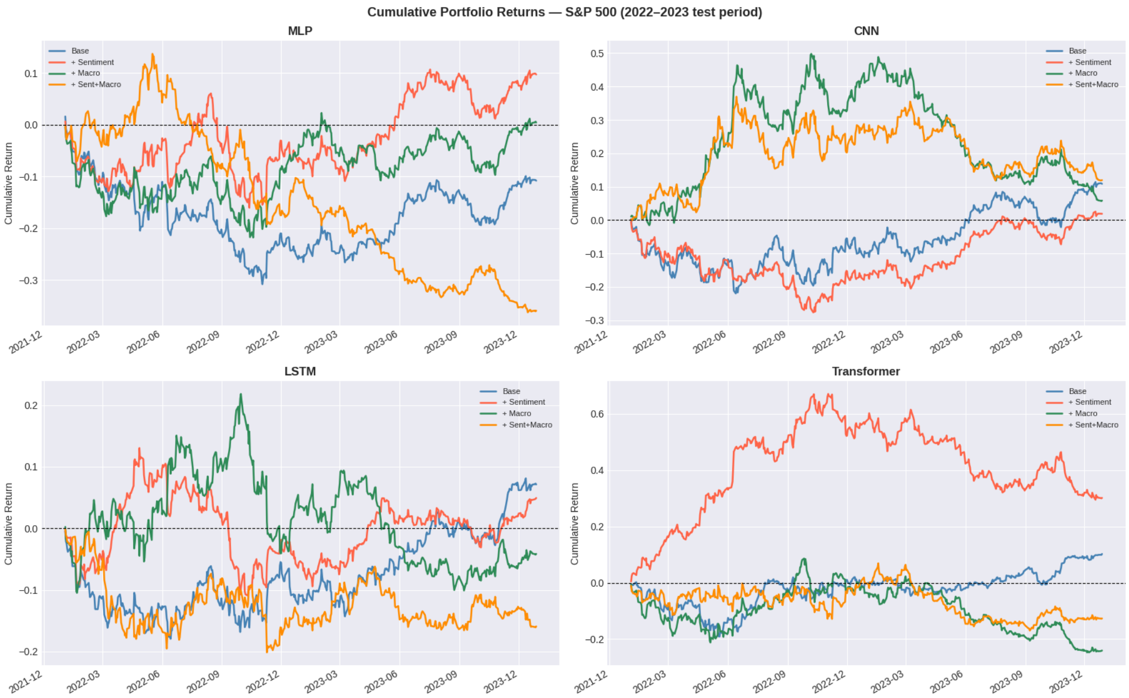


Figure 6: Cumulative Portfolio Returns - S&P 500 (2022-2023 Test Period), all four models

drawdown in the entire experiment at 18.3%, alongside a total return of 12.0%. This is the only configuration where combining sentiment and macro appears to produce additive rather than subtractive effects. The CNN architecture's use of 1D convolutional filters over the time dimension performs a form of local pattern extraction that does not require global sequence attention and may act as implicit feature selection by learning which feature channels carry consistent local structure. That property may explain why CNN tolerates a 26-feature input that degrades other architectures.

MLP with all features reached a Sharpe of -1.31 and a total return of -35.9%, the worst result in the study. LSTM dropped to a Sharpe of -0.44. Transformer improved slightly relative to macro-only (-0.27 versus -0.63) but remained firmly negative. For these three architectures, combining sentiment and macro did not produce additive gains and in the MLP case produced a result substantially worse than any single-layer configuration. The reasons behind MLP failing with full features are in line with the limitations of the architecture when dealing with high-dimensional time-series inputs. Each timestep-feature pair is treated as an independent input by the MLP after flattening, so 26 features across 10 timesteps leads to a 260-dimensional input vector. Without architectural inductive biases like convolution or attention, the model is expected to have a more difficult time optimizing as the input dimension increases, especially in a small-universe scenario where 12,490 training samples may not be enough to learn reliable weights for all 260 input positions.

4.7 Ablation Study

The ablation study tested each feature layer in isolation to separate the contribution of technical, sentiment, and macro features from their combined effect. Table IV presents the full ablation results. It is evident that Macro-only results for CNN and LSTM are the strongest single-layer results in the entire analysis. Figure 7 shows the annualized Sharpe ratios from the ablation study across four isolated feature sets and four models.

Sentiment alone produced uniformly weak results. No architecture exceeded a Sharpe of 0.10 using only FinBERT probability scores. LSTM (-0.01) and Transformer (-0.11) turned negative. This does not mean sentiment carries no information; the Transformer's

Table IV: Ablation study results by isolated feature layer and model architecture.

Feature Set	Model	Total Return	Sharpe Ratio	Max Drawdown
Technical only	**MLP**	0.0111	0.1186	0.2455
Technical only	**CNN**	-0.0026	0.1024	0.3129
Technical only	**LSTM**	0.1079	0.4517	0.1701
Technical only	**Transformer**	0.1551	0.5894	0.1670
Sentiment only	**MLP**	-0.0217	0.0436	0.2960
Sentiment only	**CNN**	-0.0026	0.1024	0.3129
Sentiment only	**LSTM**	-0.0409	-0.0117	0.2949
Sentiment only	**Transformer**	-0.0865	-0.1051	0.3390
Macro only	**MLP**	0.1023	0.3346	0.2461
Macro only	**CNN**	0.5250	1.0909	0.1959
Macro only	**LSTM**	0.6769	1.3132	0.1854
Macro only	**Transformer**	-0.0105	0.0836	0.2817
All combined	**MLP**	-0.0057	0.0613	0.2173
All combined	**CNN**	0.2586	0.7495	0.1720
All combined	**LSTM**	-0.3137	-0.9916	0.3356
All combined	**Transformer**	-0.0208	0.0330	0.2169

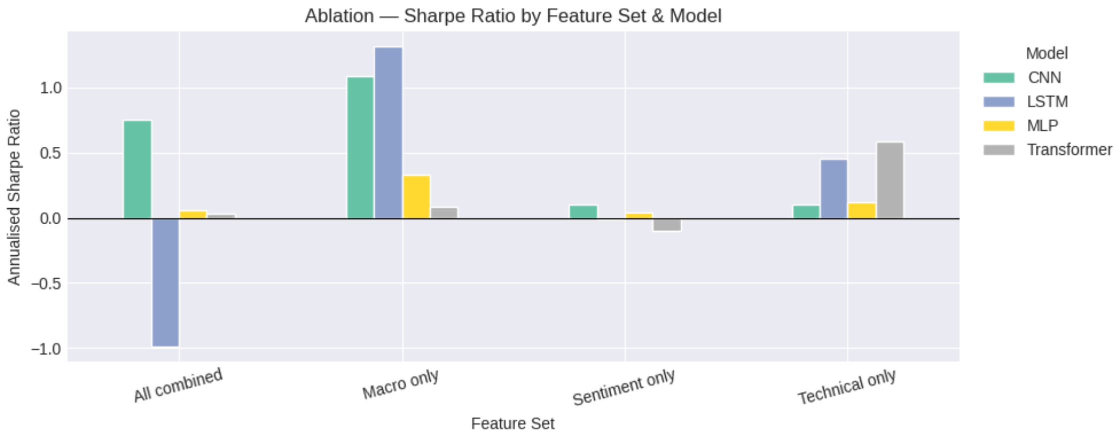


Figure 7: Ablation Sharpe Ratio by Feature Set and Model

substantial gain when sentiment was added to the technical baseline shows that sentiment scores are complementary to return history rather than independently predictive. As a standalone signal averaged across a 10-day window on a small ticker universe, the scores lack enough directional content to drive a daily rebalancing strategy. Macro alone was the strongest single-layer result for CNN and LSTM. CNN reached a Sharpe of 1.09 and LSTM reached 1.31 under macro-only features, both substantially higher than their respective technical baselines of 0.36 and 0.30. These figures are in the range of results reported by Yang *et al.*, (2025), who achieved an annualized Sharpe of 1.84 on S&P 500 data using a Temporal Fusion Transformer with an adaptive Sharpe ratio optimization approach. The architectures, test periods, and universes differ materially, but the order of magnitude is comparable.

The macro-only result for LSTM (Sharpe 1.31, total return 67.7%) is the single strongest result in the entire experiment. It occurred in isolation, using only the 12 macro-derived features without any stock-level technical information. That outcome should be interpreted with caution. It reflects one random seed, one two-year test window, and a 10-stock universe biased toward large-cap technology names that are relatively sensitive to interest rate expectations. The result is plausible given the regime. The 2022-2023 period was defined by macro-driven repricing, and LSTM's gated architecture may have learned to associate rising yield and VIX dynamics with the directional short positions that were most profitable during that specific episode. Whether that relationship generalizes to other market regimes or broader universes is not testable within this study.

Technical-only Sharpe ratios in the ablation differed from the main experiment values for several architectures. LSTM posted 0.45 in ablation versus 0.30 in the main experiment, and Transformer posted 0.59 in ablation versus 0.38. These differences arise because the ablation study is a separate model training run with independent random initialization. With 10 stocks over a two-year test window (approximately 500 trading days), single-run results are sensitive to training dynamics. The research proposal correctly identified this as a limitation as the Jobson-Korkie test would have low power in a sample of this size, and the point estimates here should be treated as directional rather than definitive.

4.8 Architecture-Feature Interaction Summary

Across all configurations, no single architecture dominated universally, and no single feature set improved all architectures. The interactions are systematic enough to draw some tentative conclusions. CNN was the most consistently robust architecture under feature augmentation. It improved or held broadly stable across three of the four feature set combinations and produced the best full-combination result (Sharpe 0.41). Its max drawdown was consistently among the lowest in the study, reaching 18.3% with all features combined. The convolutional filter mechanism appears to provide a degree of noise tolerance when input dimensionality grows.

The Transformer was the most sensitive architecture to feature set choice. It produced the highest single Sharpe in the experiment (+Sentiment: 0.96) and the sharpest single degradation (+Macro: -0.63). The attention mechanism's global receptive field may make it more responsive to signal when the right features are present and more susceptible to degradation when noisy features occupy a large portion of the input space. Kisiel and Gorse (2022) noted that the Portfolio Transformer architecture demonstrated more stable rolling Sharpe behavior than LSTM during the COVID-19 volatility period, but that study used direct Sharpe optimization rather than supervised return prediction. The supervised MSE objective used here does not discriminate between features that help predict returns and features that add volatility to predictions, which likely contributes to the Transformer's instability under macro augmentation.

LSTM showed the largest single gain from macro features in isolation (ablation Sharpe 1.31) but the steepest degradation when macro was combined with sentiment in the full feature set (Sharpe -0.44). This suggests LSTM learns macro-specific patterns effectively when those patterns are not competing with sentiment features during training but struggles to arbitrate between the two signal types within a single weight update cycle. MLP was consistently the weakest architecture across configurations. Its technical-only baseline was the only negative Sharpe among baseline results, and it posted the worst result in the experiment under full features. Short of architectural changes or a substantially larger training set, the MLP is not a viable candidate for this feature-augmentation approach.

4.9 Practical Interpretation and Limitations

Based on the results from the study, it is apparent that macro features and sentiment features had limited impact on improving the performance of all architectures uniformly while their combination did not reliably produce additive Sharpe gains. If one is selecting a single configuration from this study, the CNN with the full feature set produced the best risk-adjusted return and the lowest max drawdown (Sharpe 0.41, max drawdown 18.3%, total return 12.0%) with evidence of the most consistent behavior across augmentation types. The Transformer with sentiment augmentation produced the highest raw Sharpe (0.96), but that figure carries more variance given the architecture's sensitivity to feature set composition. The macro-only ablation result for LSTM and CNN provided the most practically actionable signal. During a regime defined by sharp, policy-driven repricing, macro features alone outperformed every other feature configuration for those two architectures. The ablation results support that interpretation for CNN and LSTM, however they do not support it for the Transformer or MLP.

Several limitations bear directly on the generalizability of these findings including survivorship bias, out-of-sample period selected and transaction costs. The current study universe was drawn from the current S&P 500 constituent list. Any stock delisted or removed between 2017 and 2023 due to bankruptcy, merger, or sustained underperformance is absent from both training and testing sets. That absence removes the worst-case performance observations from the sample leading to survivorship bias. Thus, the practical effect is upward bias in all reported returns and Sharpe ratios, since the model never had to navigate the kind of catastrophic single-stock losses that a live strategy would encounter. Correcting for this requires a point-in-time constituent list, which is not freely available. Until that data is incorporated, the reported figures should be treated as an upper bound on what a real deployment of the same strategy would have achieved.

Another limitation of the study is that the test window covers 2022 and 2023, which is approximately 500 trading days. That period was defined almost entirely by one macro regime being the Federal Reserve's rate-hiking cycle. The strong macro-only results for CNN and LSTM are plausible within that regime, but the window is too short and too regime-specific to say anything about how those configurations would behave in a normal rate environment, a liquidity crisis, or a commodity-driven inflation shock. A walk-forward validation extending the evaluation across multiple distinct macro regime ideally including the 2018 rate-tightening episode, the 2020 COVID drawdown, and the 2023 soft-landing period as separate folds would give a much more reliable picture of which feature configurations hold up and which are period-specific artifacts.

Lastly, the strategy rebalances daily, taking and exiting long and short positions across ten stocks without taking into account the transaction costs incurred. In practice, a daily rebalancing short-selling strategy faces three distinct cost layers. First, bid-ask spreads on each leg, which for large-cap S&P 500 names typically run one to three basis points per side under normal market conditions, but widen significantly during volatility spikes. Second, short-selling borrowing costs, which vary by ticker availability and borrow demand as during the 2022 drawdown, some names carried elevated borrow rates that are not captured by a flat fee assumption. Third, market impact from order execution, which is small for a ten-stock universe at this scale but non-zero. The sensitivity check at five basis points per leg is a partial proxy for the first cost layer only. Annualizing five basis points per leg per day across 250 trading days produces a drag of roughly 250 basis points annually on each leg, totaling around 500 basis points on the combined long-short book. Applied to the Transformer-plus-sentiment result (annualized Sharpe 0.96, total return 30.2%), that drag reduces the net return to approximately 25% and the Sharpe to below 0.8 which is still positive, but materially lower.

5 Conclusion

The effect of macroeconomic variables and FinBERT sentiment scores on a Transformer-based long-short portfolio model tested on the 2022-2023 S&P 500 data with 16 distinct model-feature configurations showed a Sharpe spectrum ranging from -1.31 to +0.96. This was found to be a 2.27 point bar jump caused by the exact same universe, window size, and data pipeline. FinBERT sentiment Transformer was at the top, while the same Transformer reduced to macro features and fell to -0.63. Based on the study's findings, the Transformer seems to be the most feature-sensitive architecture examined. It yielded the sharpest single degradation and the highest single Sharpe of any architecture. It is evident that its global attention mechanism is susceptible to complementary signals, such as sentiment stacked on top of price history, and that high-dimensional macro inputs that convey volatility information rather than a directional return signal cause instability. The study has also been able to uncover that a CNN model was the least sensitive to feature configuration. It showed the lowest maximum drawdown of the experiment (18.3%) with the complete feature set and remained comparatively stable across three of the four feature sets. When input dimensionality increases, its local convolutional filters seem to be additional implicit feature selection a feature that the other architectures do not have. For CNN and LSTM, using macro features alone produced the best single-layer results, with both averaging Sharpe ratios above 1.0 in the ablation study. It is noteworthy that there are three possible biases in the study that have a direct impact on the current model's performance. First off, every fund in the universe consists of ten stocks, the majority of which are consumer and large-cap technology brands. Second, survivorship bias is predicated on the fund family's historical composition rather than its current composition, and third, there are no transaction costs. Additionally, noise from a search for hyperparameters may be the cause of the Transformer's highest Sharpe. The ablation study on macro features in a rate-hiking cycle is the more practical outcome. With the same 12 macro features, the CNN and LSTM both significantly outperformed the technical baseline, but the Transformer did not benefit from macro input. Future works can conduct a formal walk-forward cross-validation with significance tests to see if macro features can be helpful during a policy-driven repricing phase. Measuring the impact on one's actual sample should only be done once it has been established that macro features are reliable additions. In order to eliminate survivorship bias, future research should also concentrate on directly addressing the 10-stock constraint by moving to a much larger universe with a historical constituent list. The single seed point estimates here should be replaced with more credible intervals by running multiple random seeds with measures of the variances around the Sharpe estimates. To put it succinctly, switching from an MSE loss to a direct Sharpe optimization objective would enable the model to learn to weight macro features conditioned on portfolio-level results rather than return fidelity.

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